

Rahul Sangaramahalingam

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🌐 LinkedIn

🐙 GitHub

🌐 Portfolio

Summary

Computer Science graduate specializing in cybersecurity, mobile application development, and software architecture. Experienced in building secure, full-stack systems, ranging from AI-integrated mobile apps to authentication and forensic security tools. Four peer-reviewed publications in cryptography, digital forensics, and secure browsing. Interested in applied security research and building practical, real-world software.

Projects

HireSight Mobile

2026

React Native, TypeScript, Groq AI, Supabase, AWS

- AI-integrated mobile application built on Supabase and AWS backend infrastructure
- Uses Groq AI for intelligent processing within a cross-platform React Native app

CineCurator Mobile

2026

React Native, Expo, Trakt API, YouTube API

- Cross-platform mobile app integrating Trakt and YouTube APIs for curated content discovery

IoT Device Security Analyzer (IDSA)

Hackhustle'25 – Top 10; Internship Offered

- Real-time CVE/NVD integration with CVSS-based risk scoring for IoT and host vulnerability scanning
- USB and network discovery via ping sweeps and port scans; checks default credentials and exposed services
- Generates HTML/PDF/JSON reports mapped to NIST and ISO 27001 frameworks
- Tkinter GUI designed for accessibility to technical and non-technical users

Quantum-Resistant IAM System

IEEE, 2025

- Enterprise identity and access management system integrating RBAC, MFA, and post-quantum cryptographic primitives
- Exposed via REST APIs for integration into existing enterprise systems
- Evaluated integration of quantum-resistant mechanisms while maintaining compatibility and scalability

SecuBrowse: Modular Browser Extension for Safe Browsing

Springer, 2026

- Manifest V3 Chrome extension using the Gemini API for real-time phishing and malicious site detection
- Detects phishing, malicious sites, and social engineering via domain and script behavior analysis

Optimized Framework for Deleted File Recovery

Unpublished, 2025

- Signature-based file carving combined with a Random Forest classifier across FAT32, NTFS, EXT4
- Classifier trained to distinguish recoverable fragments from noise, improving on pure signature-matching
- Deployed as a standalone Windows application for practical forensic use

Secure File Management System

- Web-based platform in PHP and MySQL with RBAC, encrypted file storage, and user authentication
- Implemented secure upload/download, access authorization, activity monitoring, and audit logging

Industry Experience

Python Intern, Appin Technologies, Coimbatore, India

Jun – Jul 2024

- Worked on Python-based security tooling and scripting under supervision

Education

National Sun Yat-sen University (NSYSU), Kaohsiung, Taiwan

Sep 2026 – (Incoming)

M.S., Institute of Computer Science and Engineering

Kalasalingam Academy of Research and Education, India

2022 – 2026

B.Tech, Computer Science and Engineering

CGPA: 8.29/10

Coursework: Cryptography, Network Security, Data Structures & Algorithms, Cloud Computing, Machine Learning, Operating Systems

National Chung Cheng University (NCCU), Taiwan

Sep 2024 – Jan 2025

Semester Exchange Program

Overall: 87%

Courses: Computer Networks, Operating Systems, System & Software Security, Investment Analysis, Chinese for Beginners (93/100), Psychology: Neuroscience

Publications

1. **SecuBrowse**: Browser extension on Manifest V3 using the Gemini API for real-time phishing and malicious site detection via content and behavioral analysis. Springer, 2026.
2. **Quantum-Resistant IAM**: IAM architecture combining RBAC, MFA, and post-quantum cryptographic primitives, exposed via REST APIs. IEEE, 2025.
3. **Anti-Remote Disk Imaging**: Host-based defense detecting and blocking unauthorized remote disk-imaging; achieved 92% prevention rate. IEEE, 2024.
4. **Website Authenticator**: Phishing-resistant TOTP-based authentication scheme to reduce credential theft from spoofed login pages. IEEE, 2024.

Technical Skills

Security Tools: Wireshark, Nmap, Metasploit, Burp Suite, Kali Linux, OpenVAS, Nessus, Autopsy, FTK Imager
Languages: Python, JavaScript, TypeScript, Java, PHP
Web & Backend: React.js, React Native, Node.js, Express.js, REST APIs, HTML/CSS, MySQL
AI & ML: LLM Integration, Gemini API, Groq AI, Scikit-learn
Platforms: Linux, AWS (EC2, S3), Docker, Git, GitHub, VS Code, VirtualBox, Supabase
Core Areas: Cryptographic protocols, authentication systems, digital forensics, IoT security, AI security, mobile app development

Certifications

Cyber Security Fundamentals – Teachnook Cybersecurity & Privacy – NPTEL Software Testing – NPTEL
Soft Skill Development – NPTEL Linux Platform – LinkedIn Learning

Achievements

- Team Lead – Top 10 Finalist, Hackhustle'25; IoT Device Security Analyzer; led to internship offer
- Team Lead – Participant, Techfrontier'25

Languages

Tamil (Native) English (Professional working proficiency) Hindi (Conversational) Mandarin Chinese (Beginner; 93/100 in Chinese for Beginners at NCCU; 625-day Duolingo streak)

References

Dr. K. Venkatesh

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